

Quantum complexity analysis of Time-Domain Maxwell's equations

Classiq Technologies

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Prepared by: Classiq Technologies

Summary

This document provides a scaling-level (order-of-magnitude) complexity analysis for simulating semi-discrete time-domain Maxwell dynamics and estimating a *localized* electromagnetic observable. We compare (i) a classical explicit FDTD baseline on a d -dimensional grid and (ii) quantum approaches based on block-encoding and state-of-the-art simulation primitives. The quantum cost model is explicitly end-to-end: we separate *state preparation* from *observable estimation* (readout), since readout can dominate the total cost for local observables. Two regimes are analyzed:

- **Homogeneous Maxwell (no source):** Under standard lossless assumptions, the evolution is norm-preserving. In this setting, quantum time-evolution via generalized quantum signal processing (GQSP) combined with Quantum Amplitude Estimation (QAE) can yield a polynomial improvement in the mesh parameter N for *local* observable estimation, subject to the stated (delocalization) assumptions on the solution support.
- **Driven Maxwell (time-harmonic source):** We embed the forcing into an augmented homogeneous linear system and apply Linear Combinations of Hamiltonian Simulation (LCHS). After optimizing the embedding/coupling parameter, the resulting end-to-end scaling typically exhibits a *quadratic* dependence on the evolution time T relative to the classical linear-in- T baseline, so asymptotic advantage is generally restricted to special parameter regimes and can be sensitive to assumptions on source/observable support and time horizon. A practical route to regain advantage is a hybrid workflow: drive the system only for a finite activation time, then propagate the resulting field configuration using homogeneous evolution methods.

All results in this document are scaling laws; polylogarithmic factors in ϵ and N are suppressed unless explicitly shown.

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1 Introduction

We analyze the computational scaling of simulating semi-discrete Maxwell dynamics on a d -dimensional grid with N points per dimension, with the explicit goal of estimating a *localized* electromagnetic observable supported on a subset of field degrees of freedom. The analysis is intended to be practical: we track the dominant scaling with respect to the mesh resolution N , evolution time T , and accuracy target ϵ , and we make all modeling assumptions explicit.

We benchmark against an explicit FDTD baseline [1] and analyze quantum implementations in two categories: (i) homogeneous norm-preserving dynamics simulated via generalized quantum signal processing (GQSP) [2] and (ii) time-harmonic forcing treated using LCHS [3, 4]. Crucially, we account for the readout overhead required to estimate the desired observable from the quantum state using QAE [5]. This separation between state preparation and readout is essential for realistic end-to-end complexity estimation [6].

2 Model, discretization, and notation

2.1 Discretization, degrees of freedom, and parameters

We consider a d -dimensional physical domain of linear size L , discretized using N grid points per dimension. The total number of degrees of freedom, therefore, scales as $\text{\#dof} \sim O(N^d)$.

We assume wave speed c and a total simulated evolution time T . We target a solution accuracy ϵ , measured relative to the exact solution of the *discretized* system (i.e., we treat the discretization as fixed and analyze the complexity of solving the resulting finite-dimensional evolution problem).

A time-harmonic current source $\mathbf{J}_0$ with frequency ω is supported on M_J grid points. We use ℓ to denote a characteristic geometric length scale associated with the source, and we write the typical magnitude as $\mathbf{J}_0 \sim O(J_0)$ to indicate that only order-of-magnitude dependence on J_0 is retained.

2.2 Maxwell system with a time-harmonic source

We write Maxwell's equations with a current source in schematic block form

$$\frac{d}{dt} \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} = c \begin{pmatrix} 0 & \nabla \times \\ -\nabla \times & 0 \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} + \begin{pmatrix} -\mathbf{J}_0 \sin(\omega t) \\ 0 \end{pmatrix}. \quad (1)$$

Here $\mathbf{E}$ and $\mathbf{H}$ denote discretized electric and magnetic field vectors. For the scaling arguments below, we do not require explicit material coefficients; when we invoke norm preservation, we explicitly assume a standard lossless setting with compatible boundary conditions.

2.3 Semi-discrete operator and norm scaling

Let A denote the (semi-)discrete linear operator for the homogeneous Maxwell system in first-order form, e.g., a Yee-type finite-difference time-domain (FDTD) discretization [1].

$$\frac{d}{dt} u = Au + b, \quad u = \begin{pmatrix} u_E \\ u_H \end{pmatrix}, \quad A = c \begin{pmatrix} 0 & \nabla \times \\ -\nabla \times & 0 \end{pmatrix}. \quad (2)$$

The discretized curl operators scale as the inverse grid size, i.e., $\nabla \sim (L/N)^{-1}$. Consequently, the operator norm scales as

$$\|A\| \sim O\left(\frac{c}{L/N}\right), \quad (3)$$

where $\|A\|$ is the spectral norm (largest singular value). This estimate provides the central link between spatial resolution (N) and the time-step restrictions and query complexities that appear later.

Throughout, we suppress polylogarithmic factors in ϵ and N unless explicitly retained.

3 Classical baseline: explicit FDTD complexity

We use a standard explicit FDTD scheme as a classical reference [1]. A stability condition requires a time step size of $O(\|A\|^{-1})$, i.e., the time step must scale inversely with the characteristic operator frequency scale.

If the temporal integration scheme is of order p , then achieving an error ϵ implies a step count scaling [7]

$$\text{\#steps} \sim O\left(T \|A\| \epsilon^{-1/p}\right). \quad (4)$$

Each step applies a sparse stencil to $O(N^d)$ degrees of freedom, hence costs $O(N^d)$ operations in standard implementations. Combining these contributions yields the overall classical complexity

$$O\left(\frac{c}{L} T N^{d+1} \epsilon^{-1/p}\right). \quad (5)$$

This expression is the baseline against which we evaluate potential quantum advantage. It captures the expected linear dependence on simulated time T and the mesh-dependent CFL factor through N^{d+1} .

4 Quantum cost model: state preparation vs. readout

For quantum algorithms addressing linear evolution problems, it is essential to distinguish [6]:

1. **State preparation:** prepare a normalized solution state $|u\rangle = u/\|u\|$ encoding the discrete solution vector u .
2. **Observable estimation:** estimate a desired (typically local) observable from $|u\rangle$.

This separation is not merely conceptual: in many quantum linear-dynamics algorithms, the total cost is dominated by readout when the probability mass of the solution on the measured region is small.

We focus on estimating the electromagnetic energy in a fixed local region containing M_{EH} field degrees of freedom. For such a local observable, Quantum Amplitude Estimation (QAE) [5] yields a multiplicative overhead

$$\text{\#QAE} \sim O\left(\frac{\|u\|}{\|[u]_{M_{EH}}\|} \epsilon^{-1}\right), \quad (6)$$

where $[u]_{M_{EH}}$ denotes the restriction of u to the local region. Intuitively, the ratio $\|[u]_{M_{EH}}\|/\|u\|$ quantifies how much of the solution “lives” in the region being probed; when this ratio is small, more repetitions (or amplitude-estimation iterations) are required to obtain a reliable estimate.

5 Homogeneous Maxwell: quantum complexity without a source

For the homogeneous system, we consider a state-of-the-art pipeline that (i) block-encodes $A/\|A\|$ with $\text{polylog}(N)$ overhead and (ii) performs time evolution using generalized quantum signal processing (GQSP) [2, 6]. At the level of query complexity for the block-encoding, the evolution uses

$$O(T \|A\| + \text{polylog}(\epsilon)) \quad (7)$$

queries.

Combining (7) with the norm estimate (3) yields the state-preparation complexity

$$O\left(\left(\frac{cNT}{L} + \text{polylog}(\epsilon)\right) \text{polylog}(N)\right). \quad (8)$$

This is the cost to prepare a quantum state proportional to the solution at time T (or to sample from it), up to the standard block-encoding overheads.

5.1 Norm preservation and local amplitude scaling

For lossless media and compatible boundary conditions, the homogeneous Maxwell evolution is norm-preserving in a suitable discrete energy norm [8, 1]. After normalization, we may take

$$\|u(t)\| = 1. \quad (9)$$

To estimate the QAE overhead, we require a model for how the solution weight distributes across degrees of freedom. Under the generic assumption that field amplitudes are comparable across local degrees of freedom (i.e., the solution is not strongly localized to a vanishing fraction of the grid), a typical component magnitude satisfies

$$u \sim O(N^{-d/2}), \quad (10)$$

which implies that, for a local region of size M_{EH} ,

$$\|[u]_{M_{EH}}\| \sim O(M_{EH}^{1/2} N^{-d/2}). \quad (11)$$

Substituting (9) and (11) into (6) yields

$$\#\text{QAE} \sim O(N^{d/2} M_{EH}^{-1/2} \epsilon^{-1}). \quad (12)$$

This expression makes explicit the trade-off between locality and readout difficulty: smaller measurement regions (smaller M_{EH}) generally increase QAE overhead.

5.2 End-to-end homogeneous complexity

Combining the state-preparation cost (8) with the QAE overhead (12) gives the overall end-to-end complexity

$$O\left(\frac{c}{L} T M_{EH}^{-1/2} N^{1+d/2} \text{polylog}(N) \epsilon^{-1}\right), \quad (13)$$

up to suppressed polylogarithmic factors in ϵ .

In particular, (13) yields a polynomial improvement in N over the classical baseline (5) when the goal is a local observable rather than full-state output [6]. Typically,

$$M_{EH} \sim O(N^{d_{M_{EH}}}), \quad d_{M_{EH}} \in \{0, 1, \dots, d\}, \quad (14)$$

so the scaling advantage depends on how the measurement region grows with problem size. As one illustrative case, in three-dimensional systems one obtains quartic-in- N improvement when $M_{EH} \sim O(N^3)$.

6 Driven Maxwell: quantum complexity with a time-harmonic source (LCHS)

For non-unitary (driven) dynamics, we use Linear Combinations of Hamiltonian Simulation (LCHS), which provides near-optimal dependence on problem parameters for broad classes of linear evolutions [3, 4]. In this section, we apply LCHS to a standard construction that embeds time-harmonic forcing into an augmented homogeneous linear system.

6.1 Augmented homogeneous system

We model the driven system by augmenting the state with a source oscillator so that the combined dynamics are homogeneous:

$$\frac{d}{dt}u = Au, \quad u = \begin{pmatrix} u_{EH} \\ u_J \end{pmatrix}, \quad A = \begin{pmatrix} A_{EH} & A_\Omega \\ 0 & A_J \end{pmatrix}, \quad (15)$$

where

$$A_{EH} = c \begin{pmatrix} 0 & \nabla \times \\ -\nabla \times & 0 \end{pmatrix}, \quad A_J = \begin{pmatrix} 0 & \omega \mathbb{1} \\ -\omega \mathbb{1} & 0 \end{pmatrix}, \quad A_\Omega = \begin{pmatrix} \Omega \mathbb{1} & 0 \\ 0 & 0 \end{pmatrix}. \quad (16)$$

Here, $\mathbb{1}$ are N^d -dimensional identity operators and u_J is a $2N^d$ -dimensional vector. The scalar Ω is a tunable coupling that maps the source subspace into the field subspace; it is an algorithmic parameter introduced by the embedding and will be optimized later.

By linearity, we fix the initial state on the source subspace as

$$[u(0)]_J = \mathbf{J}_0, \quad (17)$$

so that

$$\|u(0)\| \sim M_J^{1/2} \mathcal{O}(J_0). \quad (18)$$

The physical scaling is recovered by

$$\mathbf{E}(t) = \Omega^{-1} [u(t)]_E. \quad (19)$$

This relation highlights the role of Ω : increasing Ω strengthens the embedding from the source oscillator into the field variables in the augmented system, but the physical field is obtained by rescaling back by Ω^{-1} .

6.2 Symmetric/antisymmetric decomposition and the conditioning parameter

LCHS analyses use the decomposition [3, 4]

$$A = A_- + A_+, \quad A_- = \frac{A - A^\top}{2}, \quad A_+ = \frac{A + A^\top}{2}. \quad (20)$$

The eigenvalues of the symmetric part A_+ govern an exponential conditioning factor in LCHS. Define $\lambda_{\max}$ as the largest eigenvalue of A_+ . For A in (15), the symmetric part is

$$A_+ = \begin{pmatrix} 0 & \frac{1}{2} A_\Omega \\ \frac{1}{2} A_\Omega^\top & 0 \end{pmatrix}, \quad (21)$$

so

$$\lambda_{\max} = \frac{1}{2} \Omega. \quad (22)$$

Thus, the embedding parameter Ω directly controls the magnitude of the LCHS exponential factor through $\lambda_{\max} T$.

6.3 Operator-norm bound and frequency resolution

For block-encoding scaling, we use an operator-norm upper bound consistent with standard end-to-end complexity accounting [6].

$$\|A\| \sim \mathcal{O}\left(\max\left\{\frac{cN}{L}, \omega, \Omega\right\}\right). \quad (23)$$

This captures, at the scaling level, contributions from the Maxwell operator scale (cN/L), the oscillator frequency (ω), and the coupling scale (Ω).

To resolve a frequency- ω drive on a grid, we assume

$$\omega \lesssim \mathcal{O}\left(\frac{cN}{L}\right), \quad (24)$$

hence

$$\|A\| \sim \mathcal{O}\left(\max\left\{\frac{cN}{L}, \Omega\right\}\right). \quad (25)$$

This assumption places the drive within the resolvable bandwidth implied by the discretization scale.

6.4 LCHS state preparation and local observable estimation

A standard LCHS bound gives preparation complexity for the scaled state $e^{-\lambda_{\max} T} u(T)$ of [3, 4]

$$\mathcal{O}\left(\frac{\|u(0)\|}{\|e^{-\lambda_{\max} T} u(T)\|} T \|A\| \text{polylog}(\epsilon)\right). \quad (26)$$

In words: the cost is proportional to $T \|A\|$ (as expected for linear dynamics simulation) but multiplied by a conditioning ratio comparing the initial norm to the norm of the *scaled* final state, reflecting the intrinsic non-unitarity of the dynamics.

Estimating a local observable over M_{EH} degrees of freedom via amplitude estimation contributes the multiplicative cost [5]

$$\#\text{QAE} \sim \mathcal{O}\left(\frac{\|u(T)\|}{\|[u(T)]_{M_{EH}}\|} \epsilon^{-1}\right), \quad (27)$$

and hence the end-to-end complexity scales as

$$\mathcal{O}\left(e^{\lambda_{\max} T} \frac{\|u(0)\|}{\|[u(T)]_{M_{EH}}\|} T \|A\| \epsilon^{-1}\right). \quad (28)$$

The remaining task is to estimate $\|[u(T)]_{M_{EH}}\|$ in terms of the physical field amplitude produced by the source.

6.5 Field-strength scaling and detuning

We incorporate a time-harmonic source supported on M_J grid points. Away from exact resonances, the steady-state field amplitude at a typical grid point is bounded by a detuning scale Δ (distance to the nearest driven eigenfrequency), giving the schematic scaling

$$\mathbf{E}(t) \sim \mathcal{O}\left(\frac{I_0 \ell}{L^3 \Delta}\right), \quad (29)$$

where I_0 is a total-current scale.

To relate I_0 to the typical current-density amplitude J_0 , we use the dimensional relation

$$I_0 \cdot \ell \sim J_0 \times (\text{source volume}). \quad (30)$$

In a d -dimensional grid representing a 3D physical domain, each active grid site corresponds to a physical “cell volume” of size

$$V_{\text{cell}} = \frac{L^3}{N^d}, \quad (31)$$

so that a source supported on M_J sites has total volume $M_J V_{\text{cell}}$. Substituting into (30) gives

$$I_0 \ell \sim J_0 M_J \frac{L^3}{N^d}. \quad (32)$$

Substituting (32) into (29) yields

$$\mathbf{E}(t) \sim \mathcal{O}\left(\Delta^{-1} M_J N^{-d} J_0\right). \quad (33)$$

This expression makes explicit the dependence of typical field amplitude on (i) detuning (Δ^{-1}), (ii) source support size (M_J), and (iii) grid refinement (N^{-d}), under the modeling assumptions stated above.

6.6 Driven end-to-end complexity and choice of the coupling parameter

In the field sector, using (33) and (19), we obtain

$$\| [u(T)]_{EH} \| \sim O\left(M_{EH}^{1/2} \Delta^{-1} \Omega M_J N^{-d} J_0\right). \quad (34)$$

This assumes that field amplitudes are comparable across the M_{EH} local degrees of freedom, so the local norm scales like $\sqrt{M_{EH}}$ times the typical component magnitude.

Combining (28), (18), (34), and (22) yields overall complexity

$$O\left(e^{\frac{1}{2}\Omega T} \Delta \Omega^{-1} T M_J^{-1/2} M_{EH}^{-1/2} N^d \|A\| \text{polylog}(N) \epsilon^{-1}\right), \quad (35)$$

up to suppressed polylogarithmic factors in ϵ .

The factor $e^{\frac{1}{2}\Omega T} \Omega^{-1}$ in (35) reflects a trade-off intrinsic to the embedding: increasing Ω tends to increase the exponential conditioning term, while decreasing Ω increases the physical rescaling penalty.

We therefore choose Ω to control this factor by setting

$$\Omega \sim O\left(T^{-1}\right), \quad (36)$$

so that

$$e^{\frac{1}{2}\Omega T} \sim O(1). \quad (37)$$

Under the physically relevant condition $T \gtrsim O\left(\frac{L}{cN}\right)$, the choice (36) implies $\Omega \lesssim O(cN/L)$ and therefore

$$\|A\| \sim O\left(\frac{cN}{L}\right). \quad (38)$$

Substituting (36) and (38) into (35) yields

$$O\left(\Delta \frac{c}{L} T^2 M_J^{-1/2} M_{EH}^{-1/2} N^{d+1} \text{polylog}(N) \epsilon^{-1}\right). \quad (39)$$

Finally, similar to (14), it is typical that

$$M_J \sim O\left(N^{d_{M_J}}\right), \quad d_{M_J} \in \{0, 1, \dots, d\}. \quad (40)$$

However, the T -dependence in (39) is quadratically worse than the classical linear scaling in (5). Consequently, even for favorable values of M_J and M_{EH} , this scaling does not yield an asymptotic quantum advantage in general, except possibly in parameter regimes where the required evolution time T is itself limited by detuning (e.g. $T \sim O((\Delta^{-1} L/c)^{1/2})$).

Most literature that argues for strong (e.g. quartic) quantum advantage in the presence of current sources often adopts a combination of assumptions such as $M_{EH} \sim O(N^d)$ (sometimes reasonable for global-energy observables), $M_J \sim O(N^d)$ (often unrealistic for physically localized sources), and $T \sim O(L/c)$ (which can implicitly restrict the dynamics to a short-time regime where multiple reflections, reverberation, or long-lived resonances do not develop) [6]. Because the driven (LCHS-based) complexity is sensitive to both the required evolution time and the effective support of the source and observable, these assumptions can materially change the scaling conclusion.

A potential route to regain quantum advantage is to *separate the forcing window from the subsequent field evolution*: apply the source only over a finite activation time (using the driven construction), then continue with homogeneous Maxwell evolution of the resulting field configuration. In this hybrid strategy, the expensive driven component is confined to the (possibly short) source-activation interval, while the longer post-activation propagation can leverage homogeneous simulation methods such as GQSP [3, 4, 2].

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